

## 2010 'A' Level H2 Mathematics Paper 1 Suggested Solutions

| Qn 1 | Suggested Solutions                                                                                                                                                                                                                                                                                                                                                              |
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| (i)  | <p>Since <math> a  =  b </math>,</p> $\sqrt{(2p)^2 + (3p)^2 + (6p)^2} = \sqrt{1^2 + 2^2 + 2^2}$ $4p^2 + 9p^2 + 36p^2 = 9$ $p^2 = \frac{9}{49}$ $p = \pm \frac{3}{7}$ <p>Note that there are 2 possible values of <math>p</math> when you solve <math>p^2 = \frac{9}{49}</math>.</p> <p>Since <math>p &gt; 0</math>, <math>p = \frac{3}{7}</math> ← Always reject with reason</p> |
| (ii) | <p><math>(a+b) \cdot (a-b)</math></p> $=  a ^2 - b \cdot a + b \cdot a -  b ^2$ $=  a ^2 -  a ^2 \quad (\because  b  =  a )$ $= 0$ <p>Properties to note here:</p> <ol style="list-style-type: none"> <li><math>a \cdot a =  a ^2</math></li> <li><math>a \cdot b = b \cdot a</math></li> </ol>                                                                                  |

| Qn 2 | Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| (i)  | <p><math>e^x (1 + \sin 2x)</math></p> $= \left(1 + x + \frac{1}{2}x^2 + \dots\right) \left[1 + 2x - \frac{(2x)^3}{6} + \dots\right]$ <p>Be careful here. You need to replace <math>x</math> by <math>2x</math> in the standard series for <math>\sin x</math>.</p> $= \left(1 + x + \frac{1}{2}x^2 + \dots\right) (1 + 2x + \dots)$ $= 1 + 2x + x + 2x^2 + \frac{1}{2}x^2 + \dots$ $= 1 + 3x + \frac{5}{2}x^2 + \dots$                                                                              |
| (ii) | <p><math>\left(1 + \frac{4}{3}x\right)^n = 1 + \frac{4}{3}nx + \frac{n(n-1)}{2}\left(\frac{4}{3}x\right)^2 + \dots</math></p> <p>Comparing the coefficient of <math>2^{\text{nd}}</math> term of each series,</p> $3 = \frac{4}{3}n$ $n = \frac{9}{4}$ <p>Be careful here. You need to replace <math>x</math> by <math>\frac{4}{3}x</math> in the standard series for <math>(1+x)^n</math></p> <p><b>Note:</b> Always "1" for the standard series, also known as Binomial Theorem to be applied</p> |

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| <p>Third term in <math>e^x(1 + \sin 2x) = \frac{5}{2}x^2</math></p> <p>Third term in <math>\left(1 + \frac{4}{3}x\right)^n = \frac{n(n-1)}{2}\left(\frac{4}{3}x\right)^2</math></p> $= \frac{9\left(\frac{5}{4}\right)}{2}\left(\frac{16}{9}\right)x^2$ $= \frac{5}{2}x^2$ <p>Therefore, the third terms in each series are equal. (shown)</p> | <p>Since this is a “<b>shown</b>” question, complete presentation is required.</p> |
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| Qn 3 Suggested Solutions |                                                                                                                                    |
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| (i)                      | <p><math>u_n = S_n - S_{n-1}</math> ←</p> $= n(2n+c) - (n-1)(2n-2+c)$ $= 2n^2 + cn - (2n^2 - 2n + cn - 2n + 2 - c)$ $= 4n - 2 + c$ |

Key learning point:  
For **any** series,  
 $u_n = S_n - S_{n-1}$

| Qn 4 Suggested Solutions |                                                                                                                                                                                                                                                                              |
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| (i)                      | <p><math>x^2 - y^2 + 2xy + 4 = 0</math></p> <p>Differentiate with respect to <math>x</math>,</p> $2x - 2y \frac{dy}{dx} + 2x \frac{dy}{dx} + 2y = 0$ ← $2x + 2y = 2y \frac{dy}{dx} - 2x \frac{dy}{dx}$ $(2y - 2x) \frac{dy}{dx} = 2x + 2y$ $\frac{dy}{dx} = \frac{x+y}{y-x}$ |

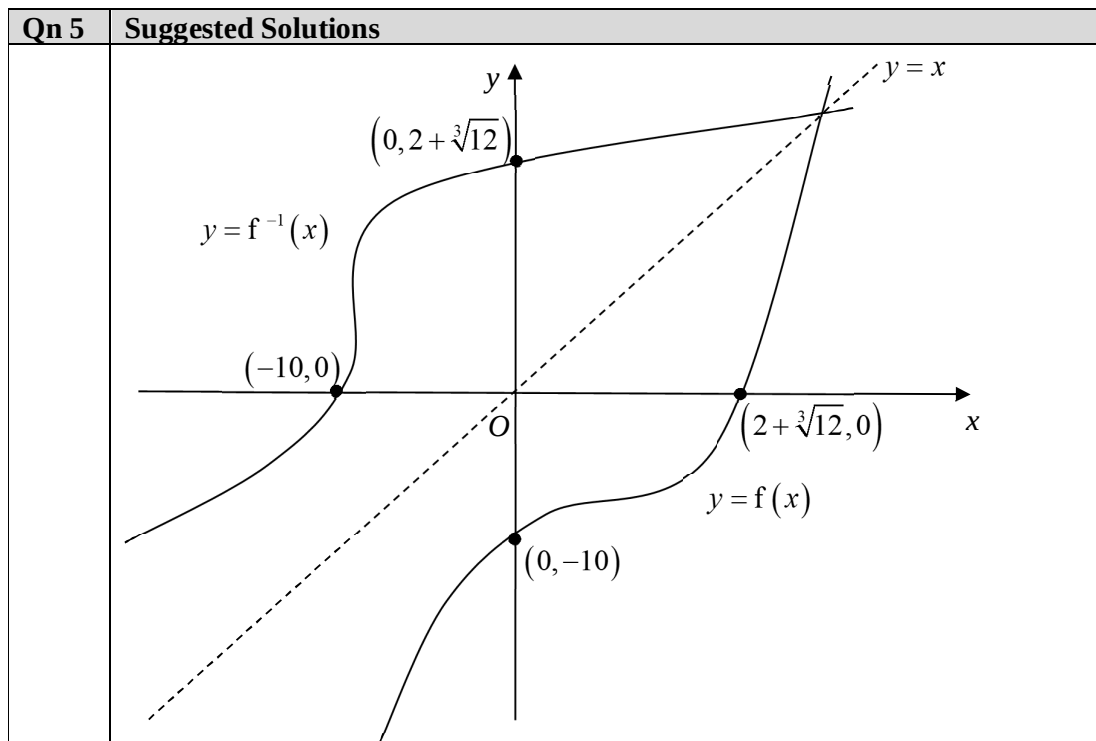
Apply concepts of implicit differentiation here.

In particular, for  $\frac{d}{dx}(2xy)$ , we will have to use product rule, i.e.  $\frac{d}{dx}(2xy) = 2x \frac{dy}{dx} + 2y$

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| (ii) | <p>For tangent parallel to the <math>x</math>-axis, <math>\frac{dy}{dx} = 0</math> ←</p> $\therefore \frac{x+y}{y-x} = 0$ $x = -y$ <p>Sub <math>x = -y</math> into equation of curve,</p> $y^2 - y^2 - 2y^2 + 4 = 0$ $y^2 = 2$ $y = \pm\sqrt{2}$ <p>When <math>y = \sqrt{2}</math>, <math>x = -\sqrt{2}</math></p> <p>When <math>y = -\sqrt{2}</math>, <math>x = \sqrt{2}</math></p> <p><math>\therefore</math> the coordinates are <math>(-\sqrt{2}, \sqrt{2})</math> and <math>(\sqrt{2}, -\sqrt{2})</math>.</p> |
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**Good to know** that if tangent is parallel to the  $y$ -axis **instead**,  $\frac{dy}{dx}$  is undefined. In that case,  $y - x = 0 \Rightarrow x = y$

| Qn 5          | Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| (i) &<br>(ii) | <div data-bbox="365 275 487 340" style="border: 1px solid black; padding: 5px; margin-bottom: 10px;"><math>y = x^3</math></div> <div data-bbox="431 373 724 443" style="border: 1px solid black; padding: 5px; margin-bottom: 10px;">Replace <math>x</math> by <math>(x - 2)</math></div> <div data-bbox="365 464 547 535" style="border: 1px solid black; padding: 5px; margin-bottom: 10px;"><math>y = (x - 2)^3</math></div> <div data-bbox="402 548 418 751" style="text-align: center;">↓</div> <div data-bbox="443 548 711 690" style="border: 1px solid black; padding: 5px; margin-bottom: 10px;">Replace <math>y</math> by <math>\frac{y}{\left(\frac{1}{2}\right)}</math></div> <div data-bbox="365 764 574 861" style="border: 1px solid black; padding: 5px; margin-bottom: 10px;"><math>y = \frac{1}{2}(x - 2)^3</math></div> <div data-bbox="402 873 418 978" style="text-align: center;">↓</div> <div data-bbox="443 888 781 957" style="border: 1px solid black; padding: 5px; margin-bottom: 10px;">Replace <math>y</math> by <math>(y + 6)</math></div> <div data-bbox="357 1001 574 1201" style="border: 1px solid black; padding: 5px; margin-bottom: 10px;"> <math display="block">y + 6 = \frac{1}{2}(x - 2)^3</math> <math display="block">y = \frac{1}{2}(x - 2)^3 - 6</math> </div> <p>Resulting curve is <math>y = \frac{1}{2}(x - 2)^3 - 6</math></p> <p>When <math>x = 0, y = -10</math></p> <p>When <math>y = 0,</math></p> $(x - 2)^3 = 12$ $x = 2 + \sqrt[3]{12}$ <p><math>\therefore</math> the coordinates are <math>(0, -10)</math> and <math>(2 + \sqrt[3]{12}, 0)</math></p> <div data-bbox="860 1352 1286 1444" style="border: 1px solid black; padding: 5px; margin-top: 20px;">         Question requires coordinates to be in exact form.       </div> |

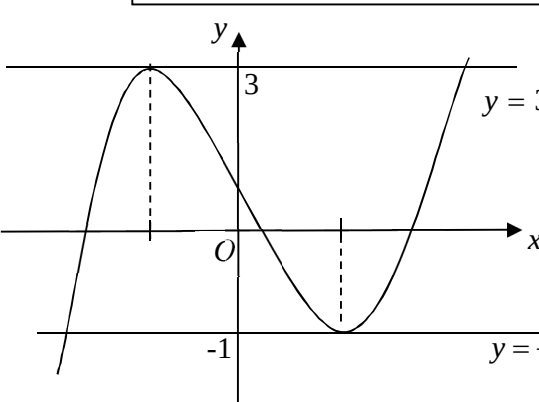


**Note:** When sketching  $y = f(x)$  and  $y = f^{-1}(x)$  on the same diagram,

1. ensure that the graphs satisfy the vertical and horizontal line test.
2. ensure that the graphs are symmetrical about the line  $y = x$ .

In addition, for this question, **axial intercepts must be in exact coordinates** as required by the question.

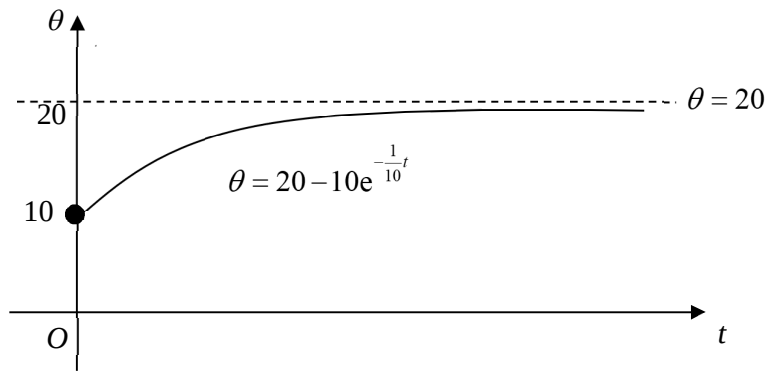
To sketch  $y = f^{-1}(x)$ , you may use **DrawInv** in GC to see the shape of  $y = f^{-1}(x)$ .

| Qn 6  | Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| (i)   | <p>Using GC,<br/> <math>\beta = 0.34730 = 0.347</math> (3 dp)<br/> <math>\gamma = 1.53209 = 1.532</math> (3 dp)</p> <div style="border: 1px solid black; padding: 5px; display: inline-block;">           Final answer needs to be in 3 decimal places as stated by the question         </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| (ii)  | <p>Required area <math>= -\int_{0.34730}^{1.53209} (x^3 - 3x + 1) dx</math><br/> <math>= 0.78142</math><br/> <math>= 0.781</math> (3 s.f.)</p> <div style="border: 1px solid black; padding: 5px; display: inline-block;"> <p>Note that <b>required area is below the x-axis</b>. You can either add a “negative” to the integral or put modulus to the integral.</p> <p>The use of GC is strongly encouraged here as question does not required exact answer.</p> <p>For the limits <math>\gamma</math> and <math>\beta</math>, it should be written in at least 5 significant figures.</p> </div>                                                                                                                                                                                                                              |
| (iii) | <p>Required area <math>= \int_{-\sqrt{3}}^0 (x^3 - 3x + 1) dx</math><br/> <math>= \left[ \frac{1}{4}x^4 - \frac{3}{2}x^2 \right]_{-\sqrt{3}}^0</math><br/> <math>= 0 - \left[ \frac{9}{4} - \frac{3}{2}(3) \right]</math><br/> <math>= \frac{9}{4}</math></p> <div style="border: 1px solid black; padding: 5px; display: inline-block;"> <p>“Use a <b>non-calculator method</b>” means that students need to show the integration steps and good presentation to arrive at the final answer.</p> <p>However, it is strongly encouraged to use the GC to check the final answer.</p> </div>                                                                                                                                                                                                                                      |
| (iv)  | <p>The number of distinct real roots for the equation <math>x^3 - 3x + 1 = k</math> is equivalent to the number of intersection points between the graphs of <math>y = x^3 - 3x + 1</math> and <math>y = k</math>.</p> <div style="border: 1px solid black; padding: 5px; display: inline-block;"> <p>Observe that <math>y = k</math> is a horizontal line</p> </div>  <p>When <math>x = -1, y = 3</math><br/>       When <math>x = 1, y = -1</math><br/> <math>\{k \in \mathbb{R} : -1 &lt; k &lt; 3\}</math></p> <div style="border: 1px solid black; padding: 5px; display: inline-block;"> <p>Set of values of <math>k</math></p> </div> <p><math>y = 3 \therefore \max k = 3</math><br/> <math>y = -1 \therefore \min k = -1</math></p> |

| Qn 7 | Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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|      | <div style="display: flex; justify-content: space-between;"> <div style="width: 45%;"> <math display="block">\frac{d\theta}{dt} = k(20 - \theta), k &gt; 0</math> <p>Given that when <math>\theta = 10</math>, <math>\frac{d\theta}{dt} = 1</math></p> <math display="block">1 = k(20 - 10)</math> <math display="block">k = \frac{1}{10}</math> <math display="block">\int \frac{1}{20 - \theta} d\theta = \frac{1}{10} \int 1 dt</math> <math display="block">-\ln 20 - \theta  = \frac{1}{10}t + C'</math> <math display="block">\ln 20 - \theta  = -\frac{1}{10}t + C, \text{ where } C = -C'</math> <math display="block"> 20 - \theta  = e^{-\frac{1}{10}t + C}</math> <math display="block">20 - \theta = \pm e^{-\frac{1}{10}t} e^C = Ae^{-\frac{1}{10}t} \text{ where } A = \pm e^C</math> <math display="block">\theta = 20 - Ae^{-\frac{1}{10}t}</math> <p>Given that when <math>t = 0</math>, <math>\theta = 10</math></p> <math display="block">\Rightarrow 10 = 20 - A \Rightarrow A = 10</math> <math display="block">\therefore \theta = 20 - 10e^{-\frac{1}{10}t} \text{ (shown)}</math> </div> <div style="width: 50%;"> <div style="border: 1px solid black; padding: 5px; margin-bottom: 10px;">             liquid is warming up           </div> <div style="border: 1px solid black; padding: 5px; margin-bottom: 10px;">             rate of increase of its temperature <math>\theta^\circ\text{C}</math> after time <math>t</math> minutes is <b>proportional</b> to the temperature difference <math>(20 - \theta)^\circ\text{C}</math> means <math>\frac{d\theta}{dt} \propto (20 - \theta) \Rightarrow \frac{d\theta}{dt} = k(20 - \theta)</math> </div> <div style="border: 1px solid black; padding: 5px; margin-bottom: 10px;">             There <b>should be a modulus sign</b> after integration.           </div> <div style="border: 1px solid black; padding: 5px; margin-bottom: 10px;">             If students want to remove this modulus, they need to provide reason like <math>20 - \theta &gt; 0</math>.           </div> <div style="border: 1px solid black; padding: 5px;">             Point to note:<br/>             Substitute the values to find the arbitrary constant only <b>after</b> the removal of the modulus from the equation.           </div> </div> </div> |
|      | $\theta = 20 - 10e^{-\frac{1}{10}t}$ <p>At <math>\theta = 15</math>,</p> $15 = 20 - 10e^{-\frac{1}{10}t}$ $10e^{-\frac{1}{10}t} = 5$ $-\frac{1}{10}t = \ln \frac{1}{2}$ $t = -10 \ln \frac{1}{2} = 10 \ln 2 = 6.93 \text{ min}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

As  $t \rightarrow \infty$ ,  $e^{-\frac{1}{10}t} \rightarrow 0$ .  $\therefore \theta \rightarrow 20$ .

i.e. the temperature of the **liquid increases and approaches** room temperature,  
i.e.  $20^\circ\text{C}$ , for large values of  $t$ .



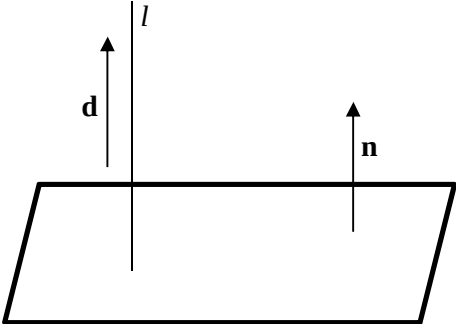
| Qn 8 | Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| (i)  | <div style="display: flex; justify-content: space-between;"> <div style="width: 45%;"> <math display="block">z_1 = 1 + i\sqrt{3}</math> <math display="block">= 2e^{\frac{\pi}{3}i}</math> <math display="block">= 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)</math> <br/> <math display="block">z_2 = -1 - i</math> <math display="block">= \sqrt{2}e^{-\frac{3\pi}{4}i}</math> <math display="block">= \sqrt{2}\left[\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)\right]</math> </div> <div style="width: 50%;"> <div style="border: 1px solid black; padding: 5px; margin-bottom: 10px;">           NORMAL FLOAT AUTO REAL RADIAN MP<br/>           MATH NUM <b>CMPLX</b> PROB FRAC<br/>           1:conj(<br/>           2:real(<br/>           3:imag(<br/>           4:angle(<br/>           5:abs(<br/>           6:►Rect<br/>           7:►Polar         </div> <div style="border: 1px solid black; padding: 5px; margin-bottom: 10px;">           NORMAL FLOAT AUTO REAL RADIAN MP<br/>           angle(1+i√3)<br/>           ..... 1.047197551<br/>           Ans/π<br/>           ..... 0.3333333333<br/>           Ans►Frac<br/>           ..... <math>\frac{1}{3}</math> </div> <div style="border: 1px solid black; padding: 5px;">           NORMAL FLOAT AUTO REAL RADIAN MP<br/>           angle(-1-i)<br/>           ..... -2.35619449<br/>           Ans/π<br/>           ..... -0.75<br/>           Ans►Frac<br/>           ..... <math>-\frac{3}{4}</math> </div> </div> </div> |
| (ii) | <div style="display: flex; justify-content: space-between;"> <div style="width: 45%;"> <math display="block">\frac{z_1}{z_2} = \frac{2e^{\frac{\pi}{3}i}}{\sqrt{2}e^{-\frac{3\pi}{4}i}}</math> <math display="block">= \sqrt{2}e^{\left(\frac{\pi}{3} + \frac{3\pi}{4}\right)i}</math> <math display="block">= \sqrt{2}e^{\frac{13\pi}{12}i}</math> <math display="block">\equiv \sqrt{2}e^{\left(\frac{13\pi}{12} - 2\pi\right)i}</math> <math display="block">= \sqrt{2}e^{\frac{11\pi}{12}i}</math> <math display="block">\left(\frac{z_1}{z_2}\right)^* = \sqrt{2}e^{\frac{11\pi}{12}i}</math> <math display="block">= \sqrt{2}\left(\cos\frac{11\pi}{12} + i\sin\frac{11\pi}{12}\right)</math> </div> <div style="width: 50%;"> <div style="border: 1px solid black; padding: 5px; margin-bottom: 10px;">           Use exponential form for simplification / evaluation.         </div> <div style="border: 1px solid black; padding: 5px; margin-bottom: 10px;">           Outside principal argument         </div> <div style="border: 1px solid black; padding: 5px;">           Recall:<br/>           1. Principal argument lies in the interval <math>(-\pi, \pi]</math><br/>           2. <math>z = re^{i\theta}</math><br/> <math>z^* = re^{i(-\theta)}</math> </div> </div> </div>                                                                                                                                                                                                                                      |



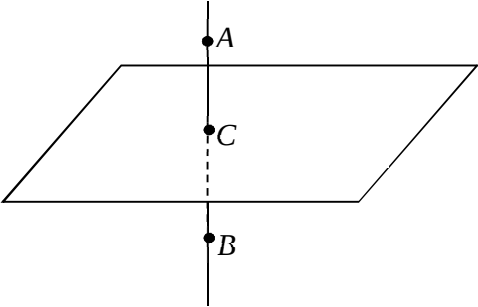
| Qn 9 | Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| (i)  | <p>Let <math>A</math> denotes the total external surface area of the box and the lid</p> $A = 3x^2 + 2(3xy) + 2xy + 3x^2 + 2(3kxy) + 2kxy$ $= 6x^2 + 8xy + 8kxy$ $= 6x^2 + 8xy(1+k)$ <div style="border: 1px solid black; padding: 5px; display: inline-block; margin-left: 100px;">       To express <math>A</math> in terms of only one variable     </div> <p>Let <math>V</math> denotes the volume of the box</p> $V = 300 = 3x^2y$ $y = \frac{100}{x^2}$ $A = 6x^2 + 8xy(1+k)$ $= 6x^2 + 8x \frac{100}{x^2} (1+k)$ $= 6x^2 + \frac{800}{x} (1+k)$ $\frac{dA}{dx} = 12x - \frac{800}{x^2} (1+k)$ $\frac{dA}{dx} = 0$ $12x = \frac{800}{x^2} (1+k)$ $x^3 = \frac{200(1+k)}{3}$ $x = \left[ \frac{200(1+k)}{3} \right]^{\frac{1}{3}}$ $\frac{d^2A}{dx^2} = 12 + \frac{1600}{x^3} (1+k)$ <p>Since <math>x &gt; 0, k &gt; 0</math></p> $\frac{d^2A}{dx^2} > 0$ <div style="border: 1px solid black; padding: 5px; display: inline-block; margin-left: 100px;">       It is a <b>must</b> to show that the area is minimum unless stated otherwise by the question. Proper reasoning is required.     </div> $x = \left[ \frac{200(1+k)}{3} \right]^{\frac{1}{3}}$ <p>such that area is minimum.</p> |

| Qn 9  | Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| (ii)  | <div style="display: flex; align-items: flex-start;"> <div style="flex: 1;"> <math display="block">\frac{y}{x} = \frac{\frac{100}{x^2}}{x}</math> <math display="block">= \frac{100}{x^3}</math> <math display="block">= \frac{100}{200(1+k)}</math> <math display="block">= \frac{3}{2(1+k)}</math> </div> <div style="flex: 1; border: 1px solid black; padding: 10px; margin-left: 10px;"> <p>Using earlier result that since<br/> <math>V = 300 = 3x^2y</math><br/> <math>y = \frac{100}{x^2}</math></p> </div> </div>                                                                                                                                                                                                                                                                                                                            |
| (iii) | <div style="display: flex; align-items: flex-start;"> <div style="flex: 1;"> <math display="block">0 &lt; k \leq 1</math> <math display="block">1 &lt; 1+k \leq 2</math> <math display="block">2 &lt; 2(1+k) \leq 4</math> <math display="block">\frac{1}{4} \leq \frac{1}{2(1+k)} &lt; \frac{1}{2}</math> <math display="block">\frac{3}{4} \leq \frac{3}{2(1+k)} &lt; \frac{3}{2}</math> <math display="block">\frac{3}{4} \leq \frac{y}{x} &lt; \frac{3}{2}</math> </div> <div style="flex: 1; border: 1px solid black; padding: 10px; margin-left: 10px;"> <p>In general, the substitution of end values to get the upper and lower bound of the inequality is <b>incorrect</b> unless certain conditions are met, i.e. the function is strictly increasing or decreasing.</p> <p>The suggested method is the preferred method.</p> </div> </div> |

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| (iv) | <p>For box to have square ends,<br/> <math>y = x</math></p> $\frac{y}{x} = \frac{3}{2(1+k)}$ $1 = \frac{3}{2(1+k)}$ $1+k = \frac{3}{2}$ $k = \frac{1}{2}$ |
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| Qn 10 | Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| (i)   | <p>If <math>l</math> is perpendicular to <math>p</math>, the direction vector of <math>l</math> is parallel to normal of <math>p</math>.</p> $\frac{x-10}{-3} = \frac{y+1}{6} = \frac{z+3}{9} (= \lambda)$ $x = 10 - 3\lambda$ $y = -1 + 6\lambda$ $z = -3 + 9\lambda$ $\mathbf{r} = \begin{pmatrix} 10 \\ -1 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 6 \\ 9 \end{pmatrix}, \quad \lambda \in \mathbb{R}$ $\text{Direction vector of } l = \begin{pmatrix} -3 \\ 6 \\ 9 \end{pmatrix}$ $= -3 \begin{pmatrix} 1 \\ -2 \\ -3 \end{pmatrix}$  <p>Since the direction vector of <math>l</math> is parallel to the normal vector of <math>p</math>,<br/> <math>l</math> is perpendicular to <math>p</math>.</p> |
| (ii)  | $l: \mathbf{r} = \begin{pmatrix} 10 \\ -1 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 6 \\ 9 \end{pmatrix}, \quad \lambda \in \mathbb{R}$ $p: \mathbf{r} \cdot \begin{pmatrix} 1 \\ -2 \\ -3 \end{pmatrix} = 0$ <p>Sub eqn (<math>l</math>) into eqn (<math>p</math>):</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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|       | $l: \mathbf{r} = \begin{pmatrix} 10 \\ -1 \\ -3 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -2 \\ -3 \end{pmatrix}, \quad \mu \in \mathbb{R}$ $\begin{pmatrix} 10 + \mu \\ -1 - 2\mu \\ -3 - 3\mu \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ -3 \end{pmatrix} = 0$ $10 + \mu + 2 + 4\mu + 9 + 9\mu = 0$ $14\mu = -21$ $\mu = -\frac{3}{2}$ $\mathbf{r} = \begin{pmatrix} 10 \\ -1 \\ -3 \end{pmatrix} + \left(-\frac{3}{2}\right) \begin{pmatrix} 1 \\ -2 \\ -3 \end{pmatrix}$ $= \begin{pmatrix} \frac{17}{2} \\ 2 \\ \frac{3}{2} \end{pmatrix}$ <p>The coordinates of point of intersection are <math>\left(\frac{17}{2}, 2, \frac{3}{2}\right)</math></p> |
| (iii) | <p>To show that point A lies on <math>l</math>,</p> $\begin{pmatrix} -2 \\ 23 \\ 33 \end{pmatrix} = \begin{pmatrix} 10 + \mu \\ -1 - 2\mu \\ -3 - 3\mu \end{pmatrix}$ $\Rightarrow -2 = 10 + \mu \Rightarrow \mu = -12$ $\Rightarrow 23 = -1 - 2\mu \Rightarrow \mu = -12$ $\Rightarrow 33 = -3 - 3\mu \Rightarrow \mu = -12$ <p>There is a consistent set of values of <math>\mu</math>. Therefore, <math>(-2, 23, 33)</math> lies on <math>l</math>.</p>                                                                                                                                                                                                    |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | $\overrightarrow{OA} = \begin{pmatrix} -2 \\ 23 \\ 33 \end{pmatrix} \quad \text{Let } \overrightarrow{OC} = \begin{pmatrix} 17/2 \\ 2 \\ 3/2 \end{pmatrix}$ <p>Using ratio theorem,</p> $\overrightarrow{OC} = \frac{\overrightarrow{OA} + \overrightarrow{OB}}{2}$ $\overrightarrow{OB} = 2 \begin{pmatrix} 17/2 \\ 2 \\ 3/2 \end{pmatrix} - \begin{pmatrix} -2 \\ 23 \\ 33 \end{pmatrix}$ $= \begin{pmatrix} 19 \\ -19 \\ -30 \end{pmatrix}$  <p>The coordinates of point B are (19, -19, -30).</p> |
| (iv) | $\text{Area of triangle } OAB = \frac{1}{2}  \overrightarrow{OA} \times \overrightarrow{OB} $ $= \frac{1}{2} \left  \begin{pmatrix} -2 \\ 23 \\ 33 \end{pmatrix} \times \begin{pmatrix} 19 \\ -19 \\ -30 \end{pmatrix} \right $ $= \frac{1}{2} \left  \begin{pmatrix} -63 \\ 567 \\ -399 \end{pmatrix} \right $ $= 348.08$ $= 348 \text{ (nearest whole number)}$                                                                                                                                                                                                                       |

| Qn 11 | Suggested Solutions                                                                                                                                                           |                                                                                 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| (i)   | $x = t + \frac{1}{t}$ $\frac{dx}{dt} = 1 - \frac{1}{t^2} = \frac{t^2 - 1}{t^2}$ $\frac{dy}{dx} = \frac{\frac{t^2 + 1}{t^2}}{\frac{t^2 - 1}{t^2}}$ $= \frac{t^2 + 1}{t^2 - 1}$ | $y = t - \frac{1}{t}$ $\frac{dy}{dt} = 1 + \frac{1}{t^2} = \frac{t^2 + 1}{t^2}$ |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                             |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <p>At point <math>P</math>, the parameter is <math>p</math>. Therefore <math>x = p + \frac{1}{p}</math>, <math>y = p - \frac{1}{p}</math>, <math>\frac{dy}{dx} = \frac{p^2 + 1}{p^2 - 1}</math></p> <p>Equation of tangent at point <math>P</math> is</p> $y - \left(p - \frac{1}{p}\right) = \frac{p^2 + 1}{p^2 - 1} \left[ x - \left(p + \frac{1}{p}\right) \right]$ $(p^2 - 1)y - (p^2 - 1)\left(\frac{p^2 - 1}{p}\right) = (p^2 + 1)x - (p^2 + 1)\frac{p^2 + 1}{p}$ $(p^2 + 1)x - (p^2 - 1)y = (p^2 + 1)\left(\frac{p^2 + 1}{p}\right) - (p^2 - 1)\left(\frac{p^2 - 1}{p}\right)$ $= \frac{1}{p} \left[ (p^2 + 1)^2 - (p^2 - 1)^2 \right]$ $= \frac{1}{p} \left[ (p^2 + 1 + p^2 - 1)(p^2 + 1 - p^2 + 1) \right]$ $= \frac{1}{p} (2p^2)(2) = 4p \text{ (shown)}$ | <div data-bbox="1068 289 1518 394"> <p>General equation of tangent is<br/> <math>y - y_0 = m(x - x_0)</math></p> </div> <div data-bbox="1157 541 1485 779"> <p>Since this is a “shown” question, students are required to show proper presentation on how they arrive at the final shown result.</p> </div> |
| (ii) | $(p^2 + 1)x - (p^2 - 1)y = 4p \text{ -----(1)}$ <p>When tangent at <math>P</math> meets the line <math>y = x</math>, substitute <math>y = x</math> into (1)</p> $(p^2 + 1)x - (p^2 - 1)x = 4p$ $x(p^2 + 1 - p^2 + 1) = 4p$ $2x = 4p$ $x = 2p$ $\therefore y = 2p$ $A(2p, 2p)$<br>$(p^2 + 1)x - (p^2 - 1)y = 4p \text{ -----(1)}$ <p>When tangent at <math>P</math> meets the line <math>y = -x</math>, substitute <math>y = -x</math> into (1)</p> $(p^2 + 1)x - (p^2 - 1)(-x) = 4p$ $x(p^2 + 1 + p^2 - 1) = 4p$ $x = \frac{2}{p}$ $\therefore y = -\frac{2}{p}$ $B\left(\frac{2}{p}, -\frac{2}{p}\right)$ <p>Note: The line <math>y = x</math> and <math>y = -x</math> are perpendicular. Thus <math>\angle AOB = 90^\circ</math></p>                              |                                                                                                                                                                                                                                                                                                             |

$$\begin{aligned}
 \text{Area of triangle } OAB &= \frac{1}{2} \times OA \times OB \\
 &= \frac{1}{2} \times \sqrt{4p^2 + 4p^2} \times \sqrt{\frac{4}{p^2} + \frac{4}{p^2}} \\
 &= \frac{1}{2} \times \sqrt{8}p \times \frac{\sqrt{8}}{p} \\
 &= 4
 \end{aligned}$$

Therefore, the area of triangle  $OAB$  is independent of  $p$ . (shown)

(iii)

$$x = t + \frac{1}{t} \quad \text{----- (1)} \quad \leftarrow$$

$$y = t - \frac{1}{t} \quad \text{----- (2)}$$

$$(1) + (2): x + y = 2t \quad \text{----- (3)}$$

$$(1) - (2): x - y = \frac{2}{t} \quad \text{----- (4)}$$

$$(3) \times (4): (x + y)(x - y) = 4$$

The cartesian equation is

$$x^2 - y^2 = 4$$

$$\frac{x^2}{2^2} - \frac{y^2}{2^2} = 1$$

To find the equation of the oblique asymptotes, let

$$\frac{x^2}{2^2} - \frac{y^2}{2^2} = 0$$

$$y^2 = x^2$$

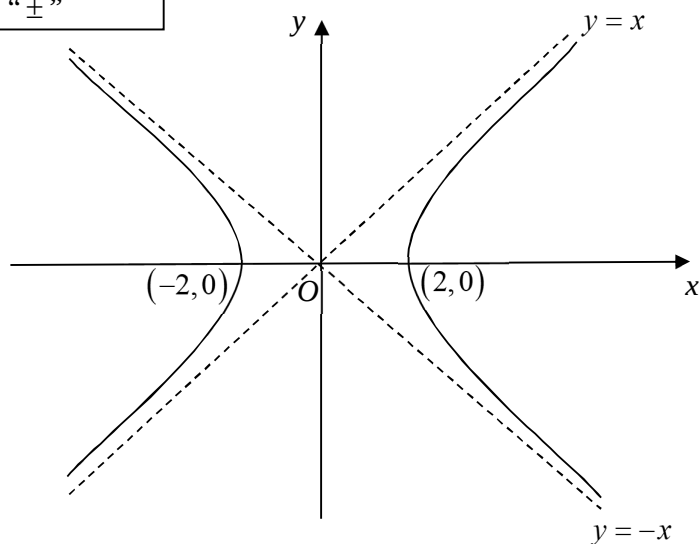
$$y = \pm x$$

When  $y = 0$ ,

$$\frac{x^2}{2^2} = 1$$

$$x = \pm 2$$

Do not forget  
the “ $\pm$ ”



The idea to obtain the cartesian equation from the parametric equations is to eliminate the parameter  $t$ .